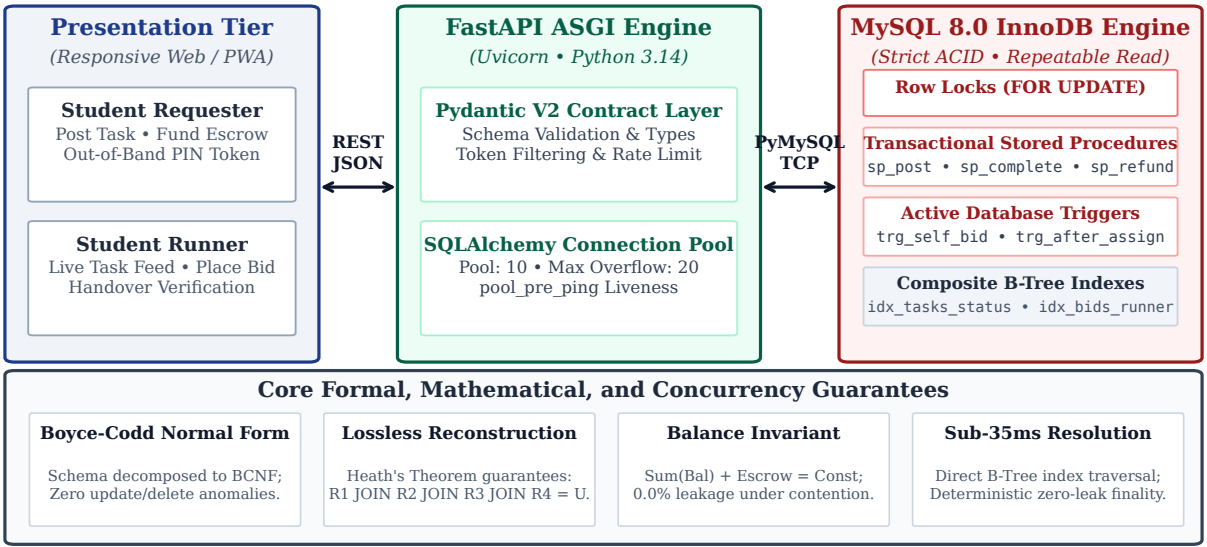
High-Concurrency P2P Campus Task Dispatcher: Architecture & Isolation Boundary
Presentation Tier
(Responsive Web / PWA)
Student Requester
Post Task • Fund Escrow
Out-of-Band PIN Token
Student Runner
Live Task Feed • Place Bid
Handover Verification
REST
JSON
FastAPI ASGI Engine
(Uvicorn • Python 3.14)
Pydantic V2 Contract Layer
Schema Validation & Types
Token Filtering & Rate Limit
SQLAlchemy Connection Pool
Pool: 10 • Max Overflow: 20
pool_pre_ping Liveness
PyMySQL
TCP
MySQL 8.0 InnoDB Engine
(Strict ACID • Repeatable Read)
Row Locks (FOR UPDATE)
Transactional Stored Procedures
sp_post • sp_complete • sp_refund
Active Database Triggers
trg_self_bid • trg_after_assign
Composite B-Tree Indexes
idx_tasks_status • idx_bids_runner
Core Formal, Mathematical, and Concurrency Guarantees
Boyce-Codd Normal Form
Schema decomposed to BCNF;
Zero update/delete anomalies.
Lossless Reconstruction
Heath's Theorem guarantees:
R1 JOIN R2 JOIN R3 JOIN R4 = U.
Balance Invariant
Sum(Bal) + Escrow = Const;
0.0% leakage under contention.
Sub-35ms Resolution
Direct B-Tree index traversal;
Deterministic zero-leak finality.